CHIPONE
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ICND2055-ICND2065-ICND2059-ICND2069
Programming Guide

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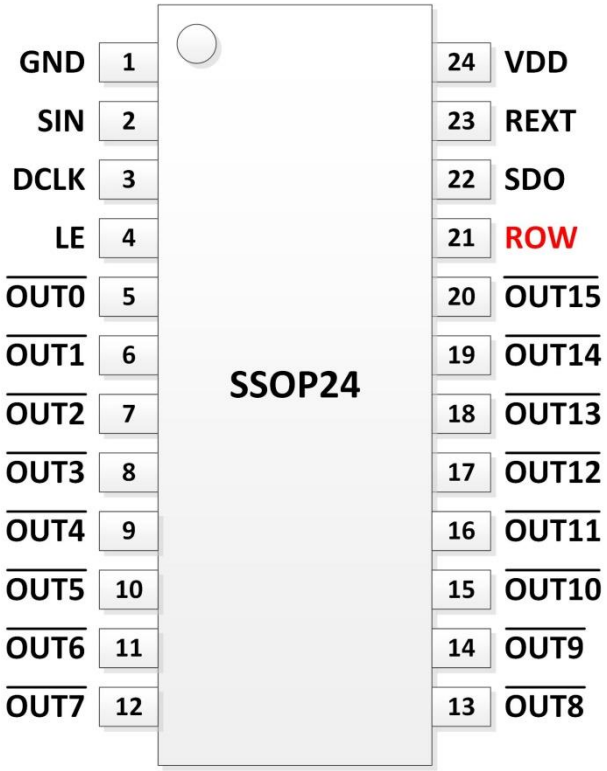
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ICND2055/ICND2065 Pin Define

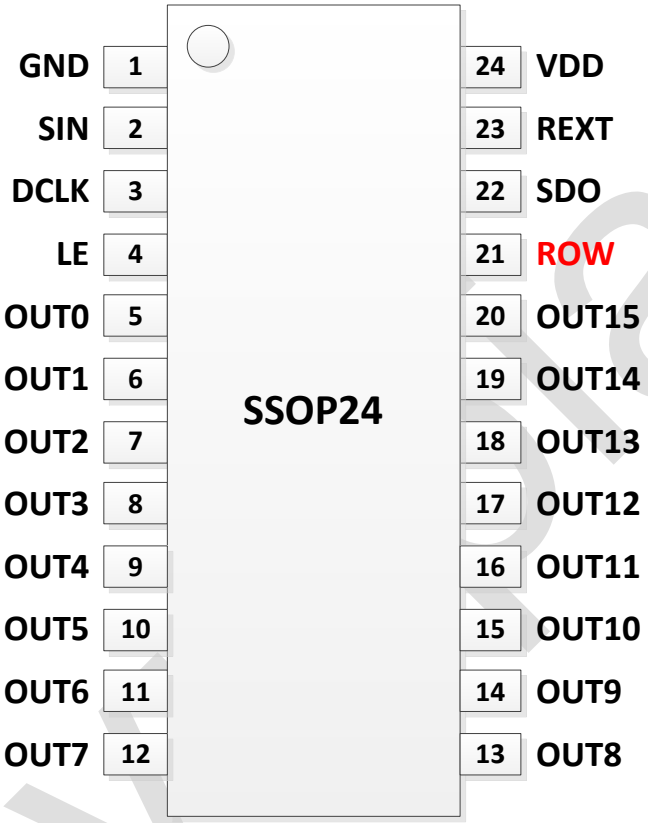
The Pin define of the ICND2055/ICND2065 is slightly different from that of the general LED display driver chip. The single CLK structure is adopted. Replace the usual GCLK signal with the ROW line signal.



ICND2055/ICND2065 (SSOP24)		
Pin No.	**Pin Name**	**Function**
1	GND	Power Ground
2	SIN	Serial data input
3	DCLK	Clock input terminal for data shift and command information
4	LE	Data transfer command input
5～20	$\overline{OUT0} \sim \overline{OUT15}$	Constant current output
21	ROW	Scan Line change signal
22	SDO	Serial data output
23	REXT	Constant-current value setting .Connection to an external resistor to GND
24	VDD	Power-supply voltage

ICND2059/ICND2069 Pin Define

The Pin define of the ICND2059/ICND2069 is slightly different from that of the general LED display driver chip. The single CLK structure is adopted. Replace the usual GCLK signal with the ROW line signal.



ICND2059/ICND2069 (SSOP24)		
Pin No.	**Pin Name**	**Function**
1	GND	Power Ground
2	SIN	Serial data input
3	DCLK	Clock input terminal for data shift and command information
4	LE	Data transfer command input
5～20	OUT0~OUT15	Constant current output
21	ROW	Scan Line change signal
22	SDO	Serial data output
23	REXT	Constant-current value setting .Connection to an external resistor to GND
24	VDD	Power-supply voltage

Shift-Register and Command Parser

A simple 16bit shift-register is integrated. All data, such as gray scale and configuration, are latched by the shift-register.

The command parser is a counter of LE length: A different length of LE indicates a different command. Such as a 1bit LE is a “Data Latch” command which indicates that there is a gray scale written in. It will send the 16bit data on shift-register to SRAM.

Control Command

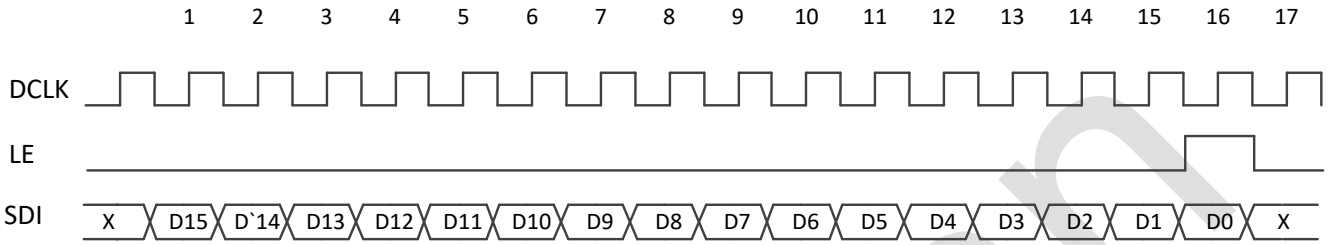
Command Name	Number of DCLK Rising Edge when LE is High	Description
DATA_LATCH	1	Transfer Serial data to buffers
VSYNC	3	Vertical Synchronal signal
WR_CFG	5	Write Configuration Register
RD_CFG	7	Read Configuration Register
SOFT_RST	9	Reset other functions except registers
EN_OP	11	Enable All Output Channels
DIS_OP	12	Disable All Output Channels
PRE_ACT	14	Pre-Active command

Note 1：The length of LE is defined as this: How many positive-edges of DCLK when LE stays logic “1”. For example, the first pulse of LE in the next figure is show a length of 1, which is a “Data Latch” command.
Note 2: After executing PRE_ACT, one or more WR_CFG or RD_CFG instructions can be executed.
Note 3: Send EN_OP first at each frame
Note 4: Once the non WR_CFG and RD_CFG instructions are received, the PRE_ACT is turned off, that is, WR_CFG or RD_CFG instruction cannot continue to receive unless PRE_ACT is sent again.
Note 5: Sending DIS_OP will close all PWM outputs until sending EN_OP instructions (waiting for 8 frames); other instructions are not limited to execution.

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Data Command

All gray scale data are latched by “Data Latch” commands. The first 16bit data, which is latched by the first “Data Latch” command, is treated as the gray scale for Line 1 Channel 15. The second 16bit data is for Line 1 Channel 14. The 17th data is for Line 2 Channel 15. The following table shows this order.



DATA LATCH

Because a ICND2055 contains 16 channel PWM output channels (OUT0~OUT15), in order to save cost, a ICND2055-ICND2065-ICND2059-ICND2069 usually controls a 16 * N LED array (in practice, N is generally 8, 16, 32, etc).One OUT channel connects N LED, working in a N row display mode when displaying.

For example, when a ICND2055 controls 16 x 8 array, it needs to send 128 DATA_LATCH instructions in one frame to write gray data for 128 LED. The data written in the first DATA_LATCH corresponds to the LED of the first line (that is, the LED of one line selected first in a frame), and the second DATA_LATCH corresponds to the LED of the first line OUT14 connection, and so on, until all the 8 rows and 128 LED are finished.

The following table is the 32 scan mode, the DATA_LATCH latch data order:

Data Order	Line	Channel
1	Line 1	Channel 15（OUT15）
2		Channel 14（OUT14）
……		……
16		Channel 0（OUT0）
17	Line 2	Channel 15（OUT15）
18		Channel 14（OUT14）
……		……
32		Channel 0（OUT0）
……		
497	Line 32	Channel 15（OUT15）
498		Channel 14（OUT14）
……		……
512		Channel 0（OUT0）

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ICND2065/ICND2069 supports 64 scan applications.

The following table is the 64 scan mode, the DATA_LATCH latch data order:

Data Order	Line	Channel
1	Line 1	Channel 15（OUT15）
2		Channel 14（OUT14）
……		……
16		Channel 0（OUT0）
17	Line 2	Channel 15（OUT15）
18		Channel 14（OUT14）
……		……
32		Channel 0（OUT0）
……		
1009	Line 64	Channel 15（OUT15）
1010		Channel 14（OUT14）
……		……
1024		Channel 0（OUT0）

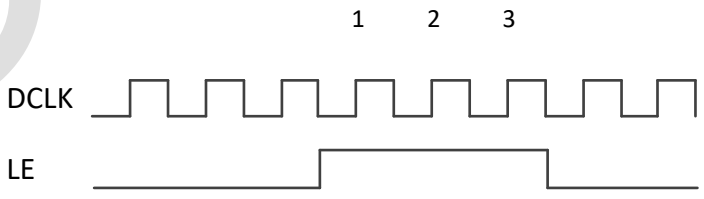
VSYNC Command

VSYNC is a frame synchronization instruction. When all data is written and displayed, it is necessary to send VSYNC instructions and control drivers to work in the next frame.

The image data of each frame is based on the VSYNC expressed by LE, that is, when the high level of the LE is detected to be 3 DCLK widths, the first data received is represented by the first valid image data of the frame, and the data is transmitted in the order of the upper section.

It should be noted that during the sending of data, other configuration commands can be inserted.

VSYNC is normally sent once per frame.



VSYNC Command

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PRE_ACT Command

The PRE_ACT instruction is used to temporarily turn off the chip write protection function. After executing PRE_ACT, one or more WR_CFG or RD_CFG instructions can be executed.

Once the non WR_CFG and RD_CFG instructions are received, the PRE_ACT is turned off, that is, WR_CFG or RD_CFG instruction cannot be continued unless the PRE_ACT is sent again.

WR_CFG Command

The WR_CFG command is used to write the register of the chip. The process of writing registers is to execute the PRE_ACT command first, and to configure the register [0x00]=8 'hAA, [0x01]=8' hAA to execute one or more WR_CFG commands; at the end, you need to configure the register [0x00]=8 'h55, [0x01]=8' h55;

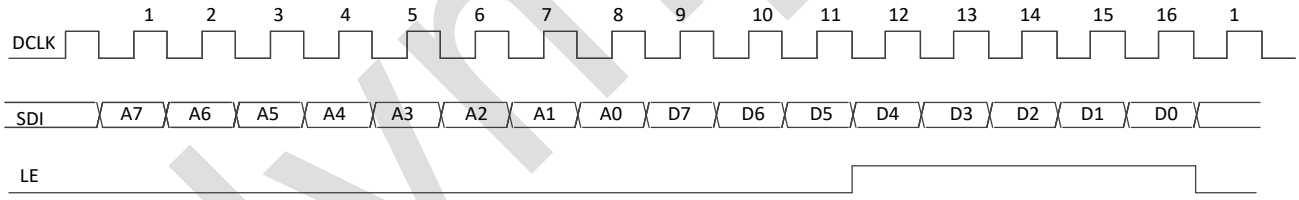
The DATA LATCH command cannot be executed during the write register, and at WR_CFG, the width of the LE is 5 DCLK, in which the first input 8bit is selected for the register address, and the later input 8bit is the selected memory value.

The following figure:

{A7, A6, A5, A4, A3, A2, A1, A0} = 8 'b1011_0101;

{D7, D6, D5, D4, D3, D2, D1, D0}=8 'b1100_1001;

That is to set the register 0xb5 (8 'b1011_0101)



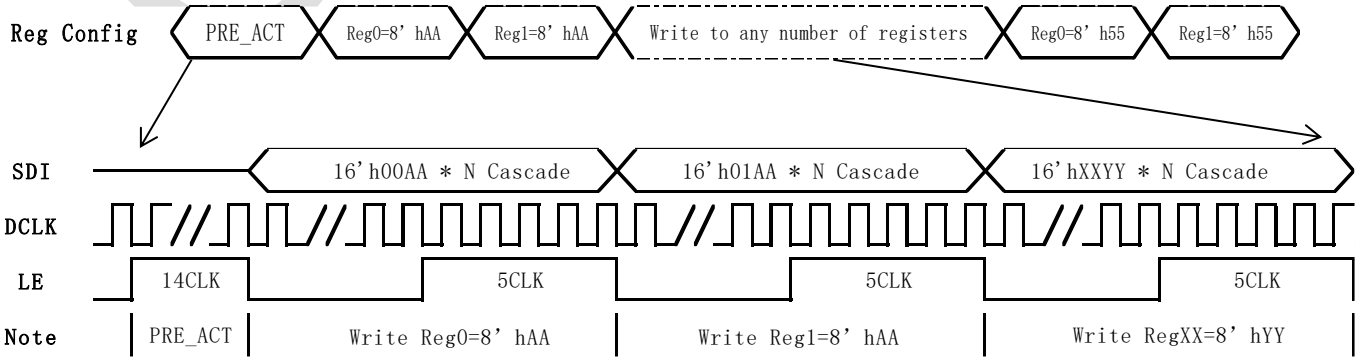
Write register timing steps:

1 Send PRE_ACTIVE instruction (LE width is 14)

2 Configuration register [0x00]=8 'hAA, [0x01]=8' hAA

3 Configure other registers (one or more)

4 Configuration register [0x00]=8 'H55, [0x01]=8' H55;



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EN_OP and DIS_OP Command

EN_OP and DIS_OP are used to open channel (OUT0~OUT15) output and close channel output respectively. These two instructions are used to prevent the display screen from displaying unexpected images when the display is just opened or when the register is changed.

After the chip is powered on, the channel state is closed by default. After the first frame data is received, the EN_OP instruction must be sent to open the channel output.

The channel state is completely controlled by these two instructions without self-opening or self-closing, that is, after the EN_OP is sent, the channel state is kept open until the DIS_OP instruction is received; after the same DIS_OP is sent, the channel state is kept closed until the EN_OP instruction is received (waiting for 8 frames).

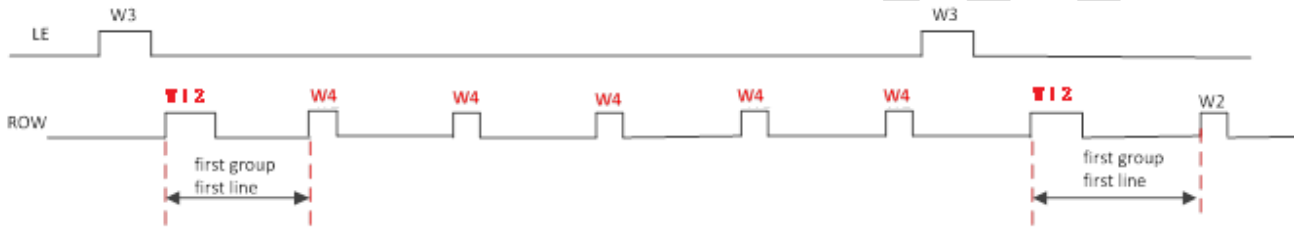
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ROW and Change Line

The rising edge of ROW indicates the beginning of a line display (synchronized with the line control signal), and the width of the high level is 12 or 4 DCLK. When the first row data of the first group (Group) is displayed, the high level width of ROW is 12 DCLK widths, and the other case is 4 DCLK widths.

The number of Row for one frame = GROUP_NUMBER * SCAN_LINES

After send the number of ROW for one frame, need send a high level width of the first line of group 1 (12 DCLK, please note that the line signal on the display module must be synchronized with the first line), and the specific grouping of the display data at this time is not concerned.



As shown in the figure above, W3 refers to the high level width of 3 DCLK, W12 refers to the high level of 12 DCLK, and W4 indicates that the high level is 4 DCLK.

When the high level of the ROW is 12 DCLK, the first line of display for the first group (Group) is to be displayed, then the next W4 is the display of the first group of second lines. I The second W12 means that the first row of the first packet of the next frame starts to show.

The number of ROW signals per frame needs to be adjusted by external control and linkage with registers.

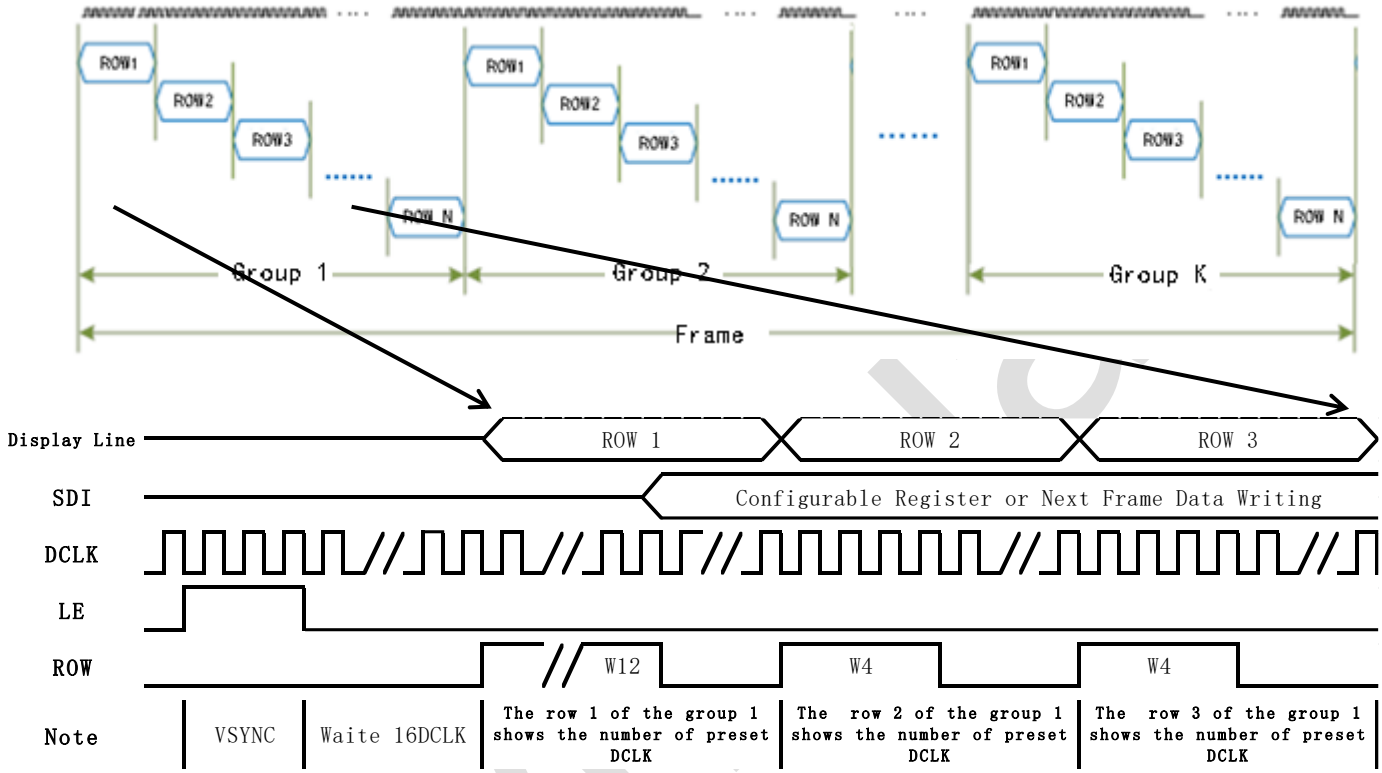
The division of each row's display time and line break time is determined by internal register configuration.

NOTE：

In the receiving card, the timing control module of the change line should use DCLK as the reference clock; each over N DCLK (the register configuration) will change the line, and produces the ROW signal.

Waveform of Display

1 After the frame starts sending VSYNC, it waits for 16 DCLK to send ROW 12 signal, displaying the first line.
2 Waiting for the number of DCLK to reach the configuration value, ROW4 signal is sent for line change.
3 Repeat step 2 until the data array is completed, waiting for the next frame to start.



NOTE：

1. The number of DCLK is configured by interface.
2. Configure Group Number, reg0x03 [6:0] = refresh rate/frame rate-1;
3. In the process of display, register configuration and data latching can be carried out simultaneously, which independent of display is timing;
4. Each frame is guaranteed to display an integer number of loops (lines 1 to n).

Software master interface description

1.DCLK Frequency

The CLK frequency is provided by the controller. It is recommended that 12.5MHz be defaulted and 4MHz < DCLK < 16MHz be used routinely.

2. Set the Gclk frequency division coefficient

Mode 1: According to the currently set DCLK frequency and different frequency dividing coefficient R, the GCLK frequency can be calculated automatically to form a drop-down frame.

Mode 2: According to the currently set DCLK frequency, different frequency division coefficient R is set to show the current GCLK frequency.

Set the Gclk frequency division coefficient R

$$R = \frac{P}{M \times N}$$

P=reg0x05[4:0] (R, G, B register synchronization)

M=reg0x04[4:0] (R, G, B register synchronization)

N=reg0x06[1:0]（(R, G, B register synchronization)

PLL related configuration and requirements as follows

1. 4MHz<FDCLK/M<10MHz（DCLK>4MHz）
2. 25MHz<FDCLK*P/M<100MHz

Gclk Frequency

GCLK =DCLK*R（R：frequency division coefficient）

When 4MHz<DCLK<8MHz：

R	P	M	N
1.75	7	1	4
2	8	1	4
2.5	10	1	4
3	12	1	4
3.5	7	1	2
4	8	1	2
4.5	9	1	2
5	10	1	2
5.5	11	1	2
6	12	1	2
7	7	1	1
8	8	1	1
9	9	1	1
10	10	1	1

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When 8MHz<DCLK<16MHz：

R	P	M	N
0.875	7	2	4
1	8	2	4
1.25	10	2	4
1.5	12	2	4
1.75	7	2	2
2	8	2	2
2.5	10	2	2
3	12	2	2
3.5	7	2	1
4	8	2	1
4.5	9	2	1
5	10	2	1
5.5	11	2	1
6	12	2	1

3. Setting Accelerate Rate

Settings can be set to 1, 2, 4, 8, 16, 32 and manual settings.

The relationship between the corresponding register settings is as follows, that is, when the main interface is set to the corresponding refresh rate, the register settings are set at the default accelerate rate of 8.

Accelerate rate 1:reg07 [7:0] = reg08 [7:0] = 255 (R, G, B register synchronization)

Accelerate ratio 2:reg07 [7:0] = reg08 [7:0] = 128

Accelerate rate 4:reg07 [7:0] = reg08 [7:0] = 64

Accelerate ratio 8:reg07 [7:0] = reg08 [7:0] = 32

Accelerate rate 16:reg07 [7:0] = reg08 [7:0] = 16

Accelerate rate 32:reg07 [7:0] = reg08 [7:0] = 8

Manual settings: reg07 [7:0]= reg08 [7:0] can be set to 1-255 arbitrary values, pop-up frame interval time (that is, the original DCLK margin) display unit is time Ms.

4. Setting Length of Line

DCKL of each Line



$$N=A+B+C+D+E=\left(\frac{0x0e[4:0]\times 2+0x7[7:0]\times 4+0x0f[3:0]\times 2+0x0d[4:0]\times 2}{R}+1\right)+E$$

A ***First line compensation Time***：0x0e[4:0]×2/R;

B ***Display Time***： $\frac{0x07[7:0]\times 4}{R}+1$;

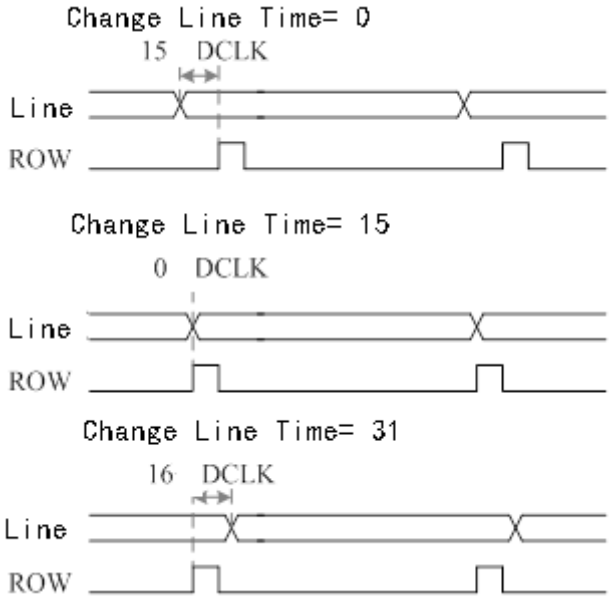
C ***Ghost reduction start Time***：0x0f[3:0] ×2/R;

D ***Ghost reduction Time***：0x0d[4:0]×2/R;

E ***Change Line Time***：The minimum value is 2.

Here we need to calculate the number of rows DCLK, N1, N2, N3 of R, G and B respectively, taking the maximum value of N1, N2 and N3 as the final result N.

Line break time position: location relation of line feed signal relative to ROW signal (0~31)



5. Refresh Rate

Refresh Rate=Group Number*Frame Rate

6. Gray level calculation formula

Grayscale = (reg0x03[6:0]+1) x (reg07[7:0]=reg08[7:0]) x 4

Set reg0x03, reg0x07, reg0x08 registers in the register. The main interface may be linked, and the grayscale of the register is configured.

if 4096<G < 8192 main interface needs to display 13bit, 8192<G less than 16384 main interface to display 14bit, 16384<G less than 32768 main interface to display 15bit, 32768<G less than 65536 main interface need to display 16bit.

7. Efficiency of brightness

Efficiency of brightness=Display Time/Total Time

Display Time of each frame= GCLK Number of each line*Line Number*Group Number/ GCLK Frequency

= (Reg07 [7:0]*4)*(Reg02 [5:0] +1)*(Reg03 [6:0] +1) /GCLK Frequency

Total Time of each frame =1/Frame Rate

Configuration interface

1. Optimization of low gray：

Drop-down option, 0~3 4 levels, R/G/B synchronization
Reg0x10[7:6], 00 corresponds to level 0;
Default R:G:B=2

2.Ghost Reduction level：

Drop-down option, 0~31 32 level R/G/B separates
Reg0x1E[4:0], 00000 corresponds to the 31 level
Default R:G:B=29:25:23.

3. .Ghost Reduction Time:

Drop-down option, 0~31 32 level R/G/B separates
Reg0x0D[4:0],00000 corresponds to level 0;
Default R:G:B=1(reg0x0D[4:0]=00001)

4. Low ash offset compensation:

Drop-down option, 0~15 16 level R/G/B separates
Reg0x23[3:0], 0000 corresponds to 0

5. Low grey first line offset compensation:

Drop-down option: 0~31 32 level R/G/B separation
Reg0x18[5:4] * 16+reg0x19[7:5],
00000 (reg0x18[5:4] at high level) corresponds to 0 level default R:G:B=8:10:16.
After the selection: 0~31 32 levels of R/G/B separation
Reg0x19[4:0] 00000 corresponds to the 0 level default R:G:B=0.
Do not reset reg0x18[5:4] * 16+reg0x19[7:5] after checking

6. Low grey first line offset time:

Drop-down option, 0~31 32 level R/G/B separates
Reg0x0E[4:0], 00000 corresponds to the 0 level
Default R:G:B=4 level (reg0x0E[4:0]=00100).

7. Coupling regulation enhancement mode:

R/G/B synchronization
Reg0x20[6] is selected as 1 defaults

8. Dark block coupling regulation

Drop-down option: 0~31 32 levels
Reg0x20[4:0], 00000 corresponds to the 0 level default R:G:B=8:8:16.

9. Coupling optimization 1:

Reg0x0C[7:0], 00000000 corresponds to 0 level, reg0x0C[7:0]< (reg0x07[7:0]=reg0x08[7:0]) x 4.
Remarks: reg0x07= reg0x08, reg0x0C[7:0] can be adjusted separately.

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10. Coupling optimization 2: ：

R/G/B separate options

Reg0x1F[5] checked for 1, R is not checked, G/B check.

11.Output current Gain：

GAIN=20%~200%，default is 100%

if GAIN >67%，reg0x1C[7]=1，reg0x1C[6:0]=63.5* GAIN

Gain=（Igain-127）*1.56% 255≥Igain≥170

if GAIN <67%，reg0x1C[7]=0，reg0x1C[6:0]=3*63.5* GAIN

Gain=Igain*0.525% 127≥Igain≥42

12. Black screen energy saving, default effect priority

Drop-down options: two modes (effect first, energy saving priority)

The effect is first: the receiving card dynamically detects each group of data channels. The current frame RGB data is all 0, the register rcg0x13[2] = 1'b0 of the current data group RGB, reg0x13[4]=1, the current frame RGB data is all 0 time reg0x13[4]=0 memo: each frame needs to send two registers, reg0x13 every frame is sent, other register order Send out

Energy saving priority: setting up reg0x13[4]=1, reg0x13[2]=1

13.Open detect：

Check to enable (reg0x16[1]=1), do not check to close this function (reg0x16[1]=0).

Open circuit detection and removal of bad points, default for power 3 seconds after automatic detection, you can manually click < remove the bad points >

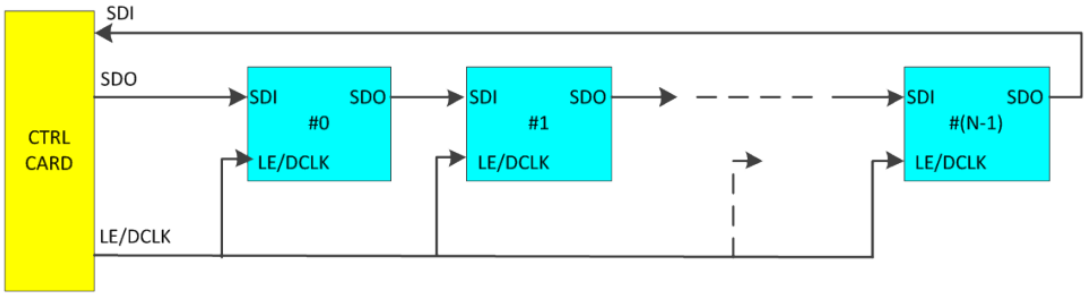
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Open detect

1. After power on 60 frames, open circuit detection is performed.
2. Send ***PRE_ACT***
3. WR reg0x00 = 8 'HAA; WR reg0x01=8' HAA; //set password
4. Configuration reg0x1C[6:0] right shift two bit, high fill 0
5. Wait for a frame of time, perform the open circuit detection according to the following steps
6. WR reg0x16 = 8 'H01; //enable open detect
7. Send Row, that is to set up Row first, then set ROW to low, high level time (recommended as 400uS) for the time required for the check line, low level time (recommended >=16 DCLK) for data acquisition and latch time. DCLK is sent continuously.
8. Line wrap, how many rows it needs to scan, and how many times the step 7 is executed.
9. After testing, Wr reg0x16 = 8 'H02; / / close open detect while enable open.
10. Alignment time is suggested to be aligned with the rising edge of ROW
11. During the opening test, LE=0.

Open detect data read back

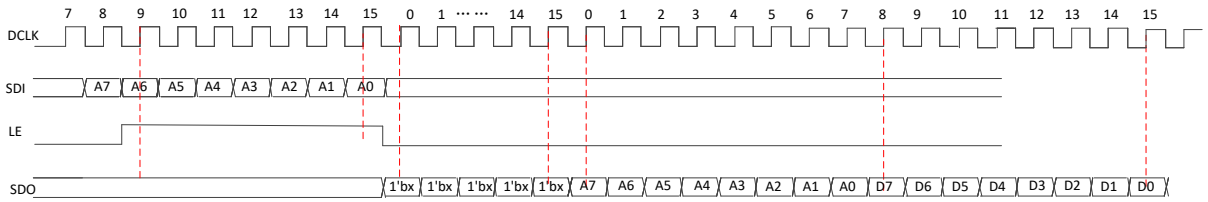
ICND2055 open detect data can be read back to the control card through SDO. The figure below is a schematic diagram of read back open detects data.



Open detect data read back is realized by register read-out function.

RD_CFG command

The ***RD_CFG*** command is used to read the registers of the chip. The process of reading the register is: SDI puts the address of the register to be read into the low 8bit, and LE contains 7 DCLKs for locking. After waiting for 16 dummy DCLKs, read out the 16bit data, of which the high 8bit is the register address and the low 8bit is the register value.

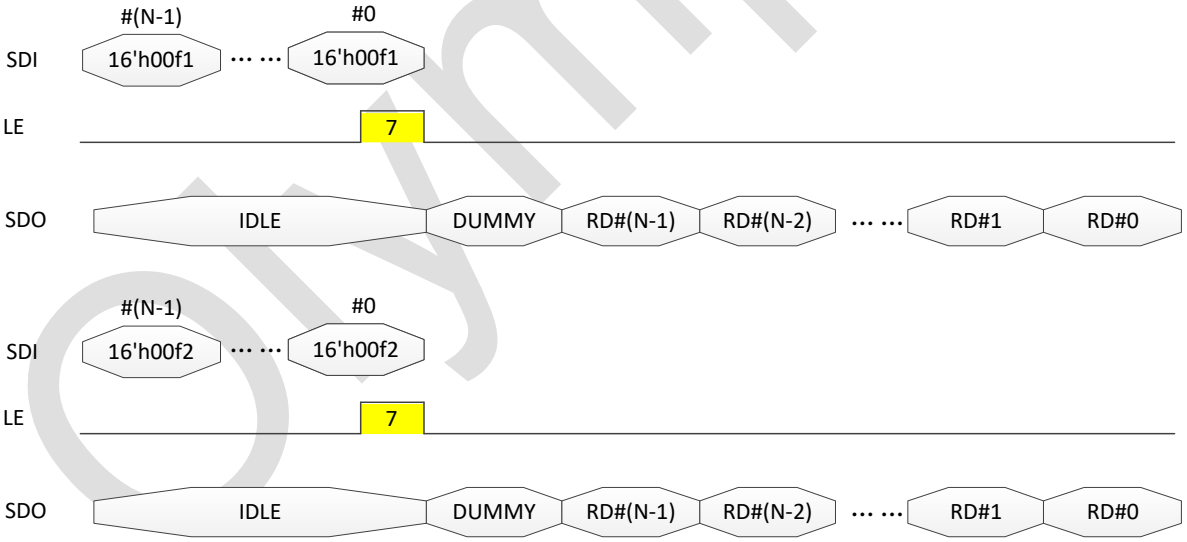


When the data of N chips need to be read out, it needs to be sent to n register addresses at a time and locked to the chip through LE at the same time. When reading, the chip at the end of the cascade reads first and then reads the chip register data forward.

Open Detect data read back sequence



1. Wait for the open detect to be completed;
2. Send ***PRE_ACT***;
3. WR reg [0x00] = 8'haa; WR reg [0x01] = 8'haa; \ \ set password
4. Set to read out the scanning line (read out one line at a time, and read out all lines through cycle setting);
WR reg [0x72] = M-1; \ \ M is the scan line to be read out, that is, when the first line needs to be read out, the register writes 0x00
WR reg [0x73] = 8'h42 (Mode A); WR reg [0x73] = 8'h82 (Mode B)
5. Read out the open circuit data. Since the read out data is 8 bit, 16 channels need to be read out twice;
The register address of CH [15:8] is 0xf1;
The register address of CH [7:0] is 0xf2;
6. Repeat steps 2-5 to read out the data of the next line. (can read a full line at a time or one line per frame)
The following figure shows the detect sequence of high-8 channel and low-8 channel respectively.



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Gamma

According to the parameter adaptive matching Gamma table given below, Set Creation provides DLL library to generate gamma table. CHIPONE provide the DLL for Gamma
Gamma table configuration needs to match parameters

INPUT_BIT_WIDTH; Set the number of bits of input data, ranging from 8 to 16
GAMMA_EXPONENT ; Gamma curvature, ranging from 0.0 to 5.0
BIT_18; 18bit Support
GROPU_NUMBER; Group number, ranging from 1 to127
WIDTH_TOTAL ; Number of DCLK each group, ranging from 0 to255

Note:
Do not optimize data for Gamma tables given by CHIPIONE.
CHIPIONE provides DLL for generating gamma tables and no longer provides gamma tables directly.
See the following section for specific instructions.

DLL

CHIPONE provide files include GammaDll_v2.2 and IcnGammaTest_v2.2
ICN2055/ICN2065/ICND2059/ICND2069 Call the following functions:
ICNGAMMA_API int GammaV2_Variable_BitWidth(int* GammaOutput, int nInputBitWidth, int nGroupNumber, int nWidthTotal, bool bIs18Bit);

Demo Tool：

1. Enter the directory :IcnGammaTest\Debug
2. Open icn_gamma.ini with notepad and modify gamma parameters:
   INPUT_BIT_WIDTH=8 ; Range 8 ~ 16，
   GAMMA_EXPONENT=0.0 ; Range 0.0 ~ 5.0, if GAMMA_EXPONENT = 0.0, use Chipone default gamma curve
   BIT_18=0 ; 0 = normal 16bit gamma curve; other = 18bit gamma curve
   GROPU_NUMBER=127 ; Range 1 ~ 127
   WIDTH_TOTAL=32 ; Range 0 ~ 255
3. Double-click IcnGammaTest.exe (or execute under the console window)
4. If executed correctly, gamma. TXT will be generated in the current directory

Note:
Both chipone_gamma.bin and IcnGamma.dll need to be placed in the same directory as executable

In order to facilitate the transplantation of automatic Gamma algorithm by the control card manufacturers, after signing NDA, the control card manufacturers can request the source code of IcnGamma.dll from CHIPONE.

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Register

The following is the default value of the register.

Name	Addr	Reg_R	Reg_G	Reg_B
Reg02	0x02	Bit[5:0]=Scan Line Number-1 Bit[7:6]=00		
Reg03	0x03	0x3f	0x3f	0x3f
Reg04	0x04	0x03	0x03	0x03
Reg05	0x05	0x0c	0x0c	0x0c
Reg06	0x06	0x02	0x02	0x02
Reg07	0x07	0x20	0x20	0x20
Reg08	0x08	0x20	0x20	0x20
Reg09	0x09	0x08	0x08	0x08
Reg10	0x0A	0x08	0x08	0x08
Reg11	0x0B	0x00	0x00	0x00
Reg12	0x0C	0x20	0x20	0x20
Reg13	0x0D	0x01	0x01	0x01
Reg14	0x0E	0x04	0x04	0x04
Reg15	0x0F	0x01	0x01	0x01
Reg16	0x10	0x82	0x82	0x82
Reg17	0x11	0x21	0x21	0x21
Reg18	0x12	0x01	0x01	0x01
Reg19	0x13	0x00	0x00	0x00
Reg20	0x14	0x00	0x00	0x00
Reg21	0x15	0x00	0x00	0x00
Reg22	0x16	0x00	0x00	0x00
Reg23	0x17	0xf0	0xf0	0xf0
Reg24	0x18	0x1f	0x1f	0x2f
Reg25	0x19	0x00	0x50	0x00

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Reg26	0x1A	0x1f	0x1f	0x1f
Reg27	0x1B	0x10	0x10	0x10
Reg28	0x1C	0xd0	0xd0	0xd0
Reg29	0x1D	0xca	0xca	0xca
Reg30	0x1E	0x42	0x46	0x48
Reg31	0x1F	0x04	0x20	0x20
Reg32	0x20	0x08	0x08	0x10
Reg33	0x21	0x01	0x01	0x01
Reg34	0x22	0x1c	0x1c	0x1c
Reg35	0x23	0x00	0x00	0x00
Reg112	0x70	0x00	0x00	0x00
Reg113	0x71	0x00	0x00	0x00
Reg114	0x72	0x00	0x00	0x00
Reg115	0x73	0x00	0x00	0x00
Reg116	0x74	0x00	0x00	0x00
Reg240	0xf0	0x00	0x00	0x00
Reg241	0xf1	0x00	0x00	0x00
Reg242	0xf2	0x00	0x00	0x00
Reg243	0xf3	0x00	0x00	0x00
Reg244	0xf4	0x00	0x00	0x00
Reg245	0xf5	0x00	0x00	0x00

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Important information

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